Supplementary Material for "Error Propagation in Cascaded Speech Translation under Acoustic Channel Degradations: A Black-Box Case Study of Commercial Speech Intelligence APIs"

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This supplementary document collects the extended numeric tables of the benchmark. All numbers are regenerated from the released API logs by the released analysis scripts. Cross-references of the form "the main text" refer to the sections of the accompanying manuscript.

S1 Complete pairwise significance results

Table S1 reports the pairwise Wilcoxon signed-rank comparisons between recognition systems that remain significant at the Holm-corrected level within the family of all pairwise comparisons of the same condition and language, so the table lists only significant results by construction. Two protocol notes are needed to read it correctly. First, comparisons among the three general-purpose recognisers are computed on the full 480-sentence main grid, 240 sentences per language, whereas every comparison involving the omni system is computed on the 144-sentence sweep subset, 72 per language, because that system never ran the full grid; the two blocks of the table are same-subset within themselves, and the omni rows state their subset explicitly. Second, the fraction of zero paired differences is substantial in mild conditions, 72 to 83 percent under reverberation and narrowband, so significance there is carried by the minority of sentences where the systems differ. The consistency of the ordering is itself informative: within every condition and language, the two LLM-ASR systems occupy the leading positions and the specialised streaming recogniser the trailing position, with the exception that Qwen3-ASR leads Qwen-Audio 3.1 in English under reverberation, clipping, narrowband, and 0 dB noise, where the refusal behaviour of the latter penalises it, while the omni system is statistically inseparable from the Qwen systems in English under noise and reverberation on its subset.

S2 Numeric results of the severity sweep

Table S2 reports the complete error rates of the eight-level severity sweep with confidence intervals, and Table S3 reports the internal frontend recogniser of the streaming translation service on the same subset. Three details of the sweep data deserve note. The confidence intervals at 0 dB and below are wide for every system, spanning ten percentage points or more, which reflects the bimodal sentence-level behaviour documented in the sentence-level variability section of the main text. The crossover between the classes occurs between 5 and 0 dB for both languages, and at 5 dB the specialised system already shows its first significant departure from its clean score while the LLM-ASR systems remain within 0.3 points of theirs in Mandarin. The internal frontend of the streaming

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Table S1: Significant Holm-corrected pairwise Wilcoxon comparisons by condition and language. Cells give the mean error rates in percent of the two systems compared. Upper block: full main grid, n=240 per language. Lower block: 144-sentence sweep subset, n=72 per language (omni system).

Condition, language	Comparison	Means (%)	p
0 dB, zh	Qwen-Audio vs Paraformer-RT	6.3 vs 41.6	< 0.001
0 dB, zh	Qwen3-ASR vs Paraformer-RT	13.1 vs 41.6	< 0.001
0 dB, zh	Qwen-Audio vs Qwen3-ASR	6.3 vs 13.1	< 0.001
0 dB, en	Qwen-Audio vs Paraformer-RT	50.2 vs 75.5	< 0.001
0 dB, en	Qwen3-ASR vs Paraformer-RT	37.1 vs 75.5	< 0.001
0 dB, en	Qwen-Audio vs Qwen3-ASR	50.2 vs 37.1	< 0.001
Reverb, zh	Qwen-Audio vs Paraformer-RT	0.2 vs 1.3	< 0.001
Reverb, zh	Qwen3-ASR vs Paraformer-RT	0.1 vs 1.3	< 0.001
Reverb, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 5.1	< 0.001
Reverb, en	Qwen-Audio vs Qwen3-ASR	1.2 vs 0.5	0.011
Narrow, zh	Qwen-Audio vs Paraformer-RT	0.1 vs 0.7	< 0.001
Narrow, zh	Qwen3-ASR vs Paraformer-RT	0.1 vs 0.7	< 0.001
Narrow, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 3.6	< 0.001
Narrow, en	Qwen-Audio vs Paraformer-RT	1.0 vs 3.6	< 0.001
Narrow, en	Qwen-Audio vs Qwen3-ASR	1.0 vs 0.5	0.015
Clip, en	Qwen3-ASR vs Paraformer-RT	0.4 vs 4.4	< 0.001
Clip, en	Qwen-Audio vs Paraformer-RT	1.6 vs 4.4	< 0.001
Clip, en	Qwen-Audio vs Qwen3-ASR	1.6 vs 0.4	< 0.001
Sweep-subset comparisons involving the omni system (n=72 per language):			
0 dB, zh	Qwen-Audio vs Omni	8.5 vs 20.2	< 0.001
0 dB, zh	Qwen3-ASR vs Omni	11.2 vs 20.2	< 0.001
0 dB, zh	Paraformer-RT vs Omni	42.4 vs 20.2	< 0.001
0 dB, en	Qwen3-ASR vs Omni	34.1 vs 43.3	0.018
0 dB, en	Paraformer-RT vs Omni	76.1 vs 43.3	< 0.001
Reverb, zh	Paraformer-RT vs Omni	1.3 vs 0.1	0.003
Reverb, en	Paraformer-RT vs Omni	5.0 vs 2.2	0.002

translation service has partial coverage in some sweep conditions, 52 to 134 of 144 sessions, because its websocket transport dropped sessions that did not recover within the retry budget; its zero score at 5 dB rests on 52 completed sessions and is a small-sample fluctuation rather than evidence of an operating point better than clean. The sub-zero conditions, 2.5, −5, and −10 dB, were run with identically seeded babble realisations, and their per-point sample sizes equal those of the original ladder, 72 sentences per language per level.

S3 Response completeness: refusals and empty transcripts

Table S4 reports the fraction of utterances for which each system's final response contains no transcript, split by the two mechanisms: protocol-level refusals, where the API returns an explicit no-words error, and silent empties, where a successful response carries an empty string. The distinction matters operationally: an explicit error can be caught by the caller and retried or escalated, whereas a silent empty return is indistinguishable from a genuinely silent audio segment. In the main grid only the 0 dB condition produces empty outputs; in the sub-zero extension they become a dominant failure mode for the streaming systems, with 42 to 58 percent of utterances returning nothing at −5 dB, while the LLM systems increasingly return hallucinated text instead. No failed

Table S2: Recognition error rates in percent on the severity sweep subset, 144 sentences per level, with bootstrap 95 percent confidence intervals. Values above 100 percent arise from insertion-dominated outputs.

System	Lang	20	15	10	5	2.5	0	−5	−10
Paraformer-v2	zh	0.8	0.8	0.8	2.5	9.8	42.4	101.2	101.9
		(0,1)	(0,1)	(0,1)	(1,4)	(6,14)	(36,50)	(98.6,104)	(99.5,104)
	en	3.8	3.7	3.9	13.4	38.1	76.1	99.5	99.8
		(2,6)	(2,6)	(2,6)	(9,18)	(31,45)	(71,81)	(98.4,101)	(99.1,101)
Qwen-Audio	zh	0.1	0.1	0.1	0.1	0.6	8.5	127.1	138.7
		(0,0)	(0,0)	(0,0)	(0,0)	(0,1)	(4,14)	(120,136)	(132,148)
	en	0.9	1.3	1.5	2.0	8.5	39.7	102.6	105.8
		(0,2)	(0,3)	(0,3)	(1,4)	(5,13)	(30,50)	(99.3,107)	(102,110)
Qwen3-ASR	zh	0.1	0.2	0.1	0.3	1.0	11.2	104.8	142.9
		(0,0)	(0,0)	(0,0)	(0,1)	(0,2)	(7,16)	(98,112)	(135,151)
	en	0.9	0.9	0.7	1.9	4.3	34.1	101.6	104.7
		(0,2)	(0,2)	(0,2)	(1,4)	(2,8)	(26,43)	(98.5,105)	(102,108)

Table S3: Error rate in percent of the streaming translation service internal frontend recogniser, with session counts; coverage is partial where the websocket transport dropped sessions.

Condition	Clean	20 dB	15 dB	10 dB	5 dB	2.5 dB	0 dB	−5 dB
Error (%)	1.0	0.8	0.2	0.9	0.0	4.0	25.4	100.8
95% CI	(0.3,1.7)	(0.3,1.6)	(0.0,0.6)	(0.3,1.6)	(0.0,0.0)	(1.9,6.7)	(19.8,31.4)	(98.4,104.1)
n	144	134	64	144	52	144	144	107
Condition	Reverb	Narrow	Codec	Loss	Clip	Tempo	−10 dB	
Error (%)	0.8	0.8	0.7	0.8	0.8	1.0	—	
95% CI	(0.3,1.5)	(0.3,1.6)	(0.2,1.4)	(0.3,1.5)	(0.3,1.6)	(0.4,1.8)	—	
n	144	144	144	144	142	142	0	

API call, of any kind, ever recovered to a different outcome on retry within the six-attempt budget, and the 24 protocol refusals of the Qwen-Audio system at 0 dB were stable across repetition runs, so the behaviour is deterministic in the input rather than an artefact of transient service conditions.

S4 Real-speech validation: complete numeric results

The three measured impulse responses used for the real-reverberation condition and the measured-RIR bridge condition are a drama room at MIT campus ($T_{60} = 0.500$ s, DRR −0.9 dB), a dining room ($T_{60} = 0.499$ s, DRR +1.3 dB), and a classroom ($T_{60} = 0.495$ s, DRR +13.1 dB); for context, the median reverberation time of the 228 survey responses that admitted reliable measurement was 0.4 s (interquartile range 0.25–0.62 s) and the median direct-to-reverberant ratio was +7.3 dB (IQR +4.1–+11.2 dB).

Table S5 reports the complete recognition results of the real-speech validation with bootstrap 95% confidence intervals over the nine conditions, together with the interpolated SNR at which each system's error crosses ten percent on the six-level real-babble ladder (values above the highest tested level are marked $> +10$; Whisper small was run only from +5 dB downward, so its crossing is marked $> +5$). Table S6 reports the corresponding translation results: the cascade under its oracle reference across all nine conditions, and the end-to-end comparison on the three conditions covered by the LiveTranslate quota. The end-to-end comparison on this reference is subject to the

Table S4: Empty final transcripts in percent of evaluated utterances, by mechanism (R: protocol refusal; S: silent empty). Main grid conditions report the full 480-sentence grid (240 per language); sub-zero conditions report the 144-sentence sweep subset (72 per language).

System	Lang	2.5 dB	0 dB	−5 dB	−10 dB	Reverb	Mechanism
Paraformer-RT-v2	zh	4.2 (S)	2.5 (S)	41.7 (S)	43.1 (S)	0	silent
	en	4.2 (S)	15.0 (S)	58.3 (S)	52.8 (S)	0	
Paraformer-RT-8K	zh	1.4 (S)	0.8 (S)	8.3 (S)	15.3 (S)	0.8 (S)	silent
Qwen-Audio-3.1	zh	0	0.8 (R)	23.6 (R)	29.2 (R)	0	refusal
	en	0	9.2 (R)	33.3 (R)	27.8 (R)	0	
Qwen3-ASR	zh/en	0	0	0	0	0	none
Qwen3.8-Omni	zh/en	0	0	n/a	n/a	0	none

same-engine affinity analysed in the main text and is interpreted there in relative-loss terms.

S5 Whisper small: complete results

Table S7 reports the complete Whisper small results on its four tested real-speech conditions: error rates with bootstrap confidence intervals, the silent-empty rate, and the Levenshtein decomposition of the error mass into insertions, deletions, and substitutions over normalised text (characters for Mandarin, words for English). The insertion fraction is the insertion count divided by the total error mass. The Mandarin insertion dominance at +5 dB (244 insertions against 83 deletions) reflects repetition-loop hallucination; English error mass remains deletion-led at every condition.

S6 Measured-RIR bridge condition: complete results

Table S8 reports recognition error under the synthetic exponentially decaying reverberation response and under the three measured survey responses at matched T_{60}, on the full 480-sentence corpus. Table S9 reports the corresponding cascade BLEU under all three references on the sweep subset (matched n=144; the 8 kHz specialist covers the Chinese half, n=72), with the synthetic-response condition in the final columns. Full-corpus measured-RIR values differ from the sweep-matched values by at most 1.3 BLEU and are included in the released analysis output.

S7 Three-reference absolute scores and paired tests

Table S10 reports the absolute clean and 0 dB BLEU scores of the five translation paths under all three references, complementing the relative-loss columns of the main translation table. Table S11 reports the paired sentence-level bootstrap comparisons between the omni path and the two LLM-frontend cascades at 0 dB on their common subset, both as absolute BLEU differences and as relative-loss differences, under each reference.

Table S12 reports the corresponding paired comparison on the real-speech validation: relative losses from clean to 0 dB real babble for the streaming end-to-end path and the Qwen-Audio cascade on their common 150-sentence subset, under the two available references, with paired sentence-level bootstrap confidence intervals (1000 resamples) for the difference. Under both references the interval crosses zero, and under the family-disjoint reference the two losses are equal to a tenth of a point.

Table S5: Real-speech validation: recognition error rate in percent (CER for Mandarin, WER for English) with bootstrap 95% confidence intervals, $n = 75$ per cell. Columns +10 to −5 are real multi-talker babble SNR levels in dB; TN denotes real transport noise at 0 dB; Cross-10% gives the interpolated babble SNR at which the error rate passes ten percent.

System	Clean	RevR	+10	+5	+2.5	0	−2.5	−5	TN	Cross-10%
Qwen-Audio zh	1.2 (0,2)	1.4 (0,3)	2.0 (1,3)	4.7 (2,8)	10.7 (5,19)	46.9 (26,72)	107.3 (81,137)	150.0 (126,177)	10.9 (4,22)	2.8
Qwen-Audio en	0.8 (0,1)	1.6 (0,3)	1.8 (1,4)	3.1 (2,5)	8.9 (6,13)	26.0 (19,34)	72.3 (64,82)	86.6 (82,91)	4.6 (3,7)	2.3
Qwen3-ASR zh	1.2 (0,2)	1.3 (0,2)	2.4 (1,4)	15.0 (6,29)	62.3 (37,92)	117.5 (86,150)	173.0 (140,210)	207.6 (180,236)	8.6 (5,13)	7.0
Qwen3-ASR en	1.2 (1,2)	1.2 (1,2)	2.8 (1,5)	5.6 (4,8)	13.7 (10,18)	24.2 (19,30)	43.5 (36,52)	64.1 (57,71)	5.4 (4,8)	3.6
Paraformer-v2 zh	4.1 (2,6)	3.2 (2,5)	12.1 (8,17)	40.1 (32,49)	64.8 (54,76)	102.6 (89,117)	131.3 (116,148)	164.4 (146,186)	35.6 (30,42)	$> +10$
Paraformer-v2 en	3.6 (2,5)	4.1 (3,5)	9.5 (7,12)	24.7 (21,28)	39.5 (35,44)	56.7 (52,62)	76.8 (72,81)	86.1 (83,89)	30.3 (25,36)	9.8
Paraformer-8K zh	5.1 (3,7)	6.3 (4,8)	27.9 (22,34)	48.6 (42,54)	60.2 (54,66)	78.7 (74,84)	86.8 (82,91)	93.1 (89,97)	77.4 (72,83)	$> +10$
Whisper small zh	10.1 (8,13)	—	—	78.7 (60,102)	—	120.3 (108,134)	—	130.7 (119,145)	—	$> +5$
Whisper small en	3.6 (2,5)	—	—	19.3 (15,24)	—	47.7 (40,55)	—	92.3 (86,99)	—	$> +5$

Table S6: Real-speech validation, translation quality: corpus BLEU of the Qwen-Audio cascade against the oracle reference (the cascade engine on human transcripts) over all nine conditions, and of the LiveTranslate end-to-end path on its three covered conditions; zh→en and en→zh are reported separately, $n = 75$ per cell.

Cascade (Qwen-Audio → qwen-mt-uni), oracle reference: Condition	zh→en	en→zh
Clean	69.2	68.9
RevR	69.7	69.8
RB+10	68.6	69.8
RB+5	63.7	68.6
RB+2.5	63.3	63.0
RB0	43.7	47.8
RB−2.5	19.0	18.1
RB−5	6.9	7.6
TN0	63.7	66.4
End-to-end (LiveTranslate), same reference: Condition	zh→en	en→zh
Clean	44.1	39.8
Reverb (real)	44.3	38.7
RB 0 dB	20.8	28.2

S8 Whisper large-v3-turbo: complete real-babble ladder

Table S13 reports the complete ladder of the locally executed Whisper large-v3-turbo model on the real-speech validation: the clean anchor, all six real-babble levels, error rates with bootstrap confidence intervals, silent-empty rates, and the error-mass decomposition over the same normalisation as every other system. The model never returns an empty transcript in any condition, in contrast to both the provider's refusal behaviour under deep noise and Whisper small's silent empties. Its Mandarin error mass shifts towards insertions as noise deepens (insertion fraction 0.08 at +10 dB against 0.27 at −2.5 dB) while its English error mass remains deletion-led at every level (deletion fraction up to 0.38), matching the failure-structure asymmetry of the provider systems and of Whisper small. The ten-percent crossing derived from these cells lies above the mildest tested level in Mandarin (14.9 percent error already at +10 dB) and at 4.4 dB in English, as discussed in Section 4.4 and Appendix D of the main paper.

Table S7: Whisper small (local execution, default decoding): error rate in percent with bootstrap 95% confidence intervals, silent-empty rate in percent, and error-mass decomposition (insertion/deletion/substitution counts and insertion fraction), $n = 75$ per cell.

Condition	Lang	Error (%, 95% CI)	Empty (%)	Ins	Del	Sub	Ins. frac.
Clean	zh	10.1 (8,13)	0.0	1	4	104	0.01
Clean	en	3.6 (2,5)	0.0	5	3	41	0.10
RB+5	zh	78.7 (60,102)	5.3	244	83	443	0.32
RB+5	en	19.3 (15,24)	0.0	35	52	170	0.14
RB0	zh	120.3 (108,134)	17.3	294	346	638	0.23
RB0	en	47.7 (40,55)	1.3	125	143	383	0.19
RB−5	zh	130.7 (119,145)	25.3	331	410	645	0.24
RB−5	en	92.3 (86,99)	5.3	255	255	821	0.19

Table S8: Recognition error rate in percent under the synthetic reverberation response (DRR −27.6 dB) and the measured survey responses (DRR −0.9 to +13.1 dB) at matched $T_{60} = 0.5$ s, full 480-sentence corpus.

System, language	Synthetic RIR	Measured RIRs
Qwen-Audio zh	0.2	0.2
Qwen-Audio en	1.2	1.1
Qwen3-ASR zh	0.1	0.2
Qwen3-ASR en	0.5	0.5
Paraformer-v2 zh	1.3	0.7
Paraformer-v2 en	5.1	3.7
Paraformer-8K zh	18.2	3.4

Table S9: Cascade BLEU under the three references (primary / oracle / GLM) on the measured-RIR bridge condition (sweep subset, n=144; 8 kHz specialist Chinese half, n=72) and under the synthetic reverberation response on the same subset. The main grid contains no 8 kHz cascade, so its synthetic columns are omitted.

	Measured RIRs			Synthetic RIR		
Frontend	primary	oracle	GLM	primary	oracle	GLM
Qwen-Audio	55.0	90.9	52.6	52.4	89.8	50.1
Qwen3-ASR	54.9	94.5	53.1	53.3	94.2	51.2
Paraformer-v2	53.3	87.0	51.8	50.7	84.4	49.7
Paraformer-8K	48.8	81.4	49.4	—	—	—

Table S10: Absolute BLEU of the five translation paths under the three references (sweep subset; cascades conditional on non-empty frontend output at 0 dB). n: cascades 144 clean, 135–144 at 0 dB; omni 84; LiveTranslate 144/132.

	Clean			Babble 0 dB		
Path	primary	oracle	GLM	primary	oracle	GLM
Cascade: Qwen-Audio+MT	53.9	89.6	51.2	44.1	68.3	42.4
Cascade: Paraformer+MT	51.5	85.6	50.6	15.8	17.4	13.8
Cascade: Qwen3-ASR+MT	53.2	94.6	52.6	43.0	66.7	40.6
E2E: Omni-Flash	68.0	48.5	60.6	45.5	34.4	42.5
E2E: LiveTranslate	59.8	56.8	55.1	50.4	44.9	45.8

Table S11: Paired bootstrap comparisons at 0 dB on the omni path's common subset ($n = 82$–84): difference (omni minus cascade) with 95% confidence intervals, under each reference. Positive relative-loss differences mean the omni path loses a larger fraction of its clean score.

Comparison	vs primary	vs oracle	vs GLM
Omni − Qwen-Audio cascade (absolute BLEU)	-1.4 (-9.4,+5.9)	-43.3 (-50.8,-34.0)	-1.7 (-10.0,+6.3)
Omni − Qwen-Audio cascade (relative-loss diff.)	+22.0 (+8.7,+32.9)	+11.7 (-5.6,+25.8)	+20.3 (+6.5,+31.7)
Omni − Qwen3-ASR cascade (absolute BLEU)	-0.2 (-7.2,+5.9)	-38.1 (-45.9,-30.5)	+0.7 (-5.9,+6.9)
Omni − Qwen3-ASR cascade (relative-loss diff.)	+21.8 (+10.5,+31.6)	+4.4 (-9.2,+18.4)	+15.4 (+2.8,+26.7)

Table S12: Real-speech paired relative-loss comparison (common subset, $n = 150$, 75 sentences per language): corpus BLEU at clean and 0 dB real babble, relative loss in percent, and the paired difference (end-to-end minus cascade) with 95% bootstrap confidence intervals. The same-engine reference is the cascade's own engine applied to the human transcripts; the family-disjoint reference is GLM applied to the same transcripts.

	LiveTranslate (E2E)			Qwen-Audio cascade			Diff.
Reference	clean	0 dB	loss%	clean	0 dB	loss%	(95% CI)
Same-engine (oracle)	42.0	24.5	41.6	69.1	45.8	33.7	$+7.9$ $(-1.5, +16.7)$
Family-disjoint (GLM)	37.3	25.1	32.6	32.9	22.1	32.7	-0.1 $(-9.0, +9.0)$

Table S13: Whisper large-v3-turbo (local execution, greedy decoding, Mandarin initial prompt): error rate in percent with bootstrap 95% confidence intervals, silent-empty rate in percent, and error-mass decomposition (insertion/deletion/substitution counts and insertion fraction) on the clean anchor and the six-level real-babble ladder, $n = 75$ per cell.

Condition	Lang	Error (%, 95% CI)	Empty (%)	Ins	Del	Sub	Ins. frac.
Clean	zh	5.9 (4,8)	0.0	7	3	52	0.11
Clean	en	2.5 (1,4)	0.0	3	4	27	0.09
RB+10	zh	14.9 (12,19)	0.0	12	16	125	0.08
RB+10	en	3.3 (2,5)	0.0	3	6	35	0.07
RB+5	zh	45.9 (36,56)	0.0	56	66	330	0.12
RB+5	en	8.5 (6,12)	0.0	12	23	74	0.11
RB+2.5	zh	76.1 (62,94)	0.0	133	171	460	0.17
RB+2.5	en	15.0 (11,20)	0.0	23	50	121	0.12
RB0	zh	105.3 (92,119)	0.0	222	202	671	0.20
RB0	en	24.8 (20,30)	0.0	46	69	218	0.14
RB−2.5	zh	136.8 (120,156)	0.0	369	288	720	0.27
RB−2.5	en	46.7 (39,55)	0.0	84	248	322	0.13
RB−5	zh	120.3 (110,132)	0.0	223	323	725	0.17
RB−5	en	70.7 (64,78)	0.0	132	344	578	0.12